

22-07-2026

Agenda: FastAPI-2

Day-1

client, server, request, response, API, IP-address, port, socket,
127.0.0.1 / localhost, 0.0.0.0, uvicorn, fastapi, api-practical
, tested via http request (chrome)

Day-2

Part-A

HTTP Request → no matter what tool you use (browser, curl, requests, postman, apidog) http request is made up of below components:

1) Method - what kind of action it is

GET → give me data

POST → here is some data, do something with it & get back the response

PUT → replace this resource with this data

DELETE → remove this resource

2) PATH → The part of the URL identifying WHAT you're asking about e.g.
'/', '/say_hello', '/say_hello/{name}', '/predict'

3) PATH parameter

name

```
def add(q):  
    do something
```

Pieces of the path itself that are variables `/items/2`

4) Query parameter

extra key=value pairs appended to the URL after a '?',
separated by '&' e.g. `/search?q=shoes&limit=5`

↑ ↑

https://www.google.com/search?q=what+is+ai&rlz=1C1CHBF_enIN1137IN1137

https://www.google.com/search?q=what+is+ai&gs_lcrp=EgZjaHJvbWUqBwgAEAAyJwlyBwgAEAAyJwlyCggBEAAyQMYgAQyCggCEAAyQMYgAQyCggDEAAyQMYgAQyBwgEEAAyAQyDQgFEAAyGwEYsQMYgAQyBwgGEAAyAQyEggHEAAyChiDARixAxiABBIKBTIHCAg

5) HEADERS

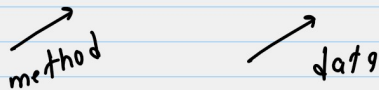
Metadata about the request, sent separately from the main content. `Content-Type : application/json` .. this tells server that the body i am sending you is JSON

6) BODY

The actual payload/data being sent, used with POST/PUT.

Data --> ML model

DATA(PUT : {sepal_length : 3.4, sepal_width:4.5}) --> Server (API) --> Prediction(sepal_length, sepal_width)



part-B

HTTP Response

1) STATUS CODE

3 digit number summarizing what happened.

2xx - success, 200 OK, 201 Created

4xx - You made a mistake (Client)

- 400 Bad Request, 404 Not Found, 422 Unprocessable Entity (Data validation fails)

5xx - The server made a mistake

- 500 Internal Server Error

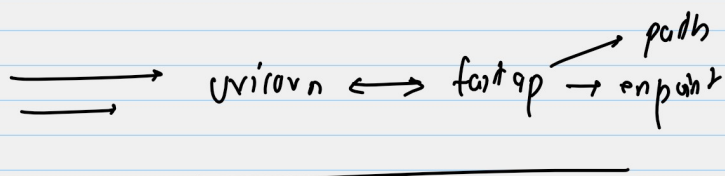
2) HEADERS

Metadata, Server is telling Client about what kind of data server is sending : `Content-Type : application/json`

3) BODY

The actual data/result being returned/requested

curl -> command line tool



`curl -X GET/post "PATH ? key=value" -H "content-type" -d "x y"`
 ↑ method (optional) ↓ default ↓ GET
 ↑ host, port, path ↑ query params
 ↑ Header
 ↑ the body

`curl -X GET -d "x y"`
 ↓ post (Turned to post)

```
@app.get("/itemsv2/{item_id}")
def read_root(item_id : int):
```

* Advantage-1 : FastAPI converts the URL Text into an actual Python Int

* Advantage-2 : If the client sends something that can't be parsed as int for e.g. /itemsv2/abc, FASTAPI automatically responds with a 422 error

item
 ↓
 { key : 1
 key
 key
 chips
 movie }

key [] : 2
 0, 1, 2

developer - API - insert data in the database

youtube -> api -> validation
 ↑
 .zip
 mp3
 []

fastapi
↑
pydantic (validation)

